

Lunar Surface Infrastructure Program WBS

Work Breakdown Structure Level 1 through Level 3 scope architecture version 0.2

Owner	Program Director	Status	Draft for review
Version	0.2	Basis	Program Charter v1.1 and WBS v0.1

Purpose and control objective

This revision decomposes the six Level 2 program elements into Level 3 deliverables. It establishes a more detailed structure for requirements allocation, scheduling, resource planning, risk ownership, performance measurement, and configuration control. The Level 3 elements remain deliverables, not schedule activities or approved work packages. Dates, budgets, control-account managers, and contractual authority are not yet assigned.

Program scope architecture

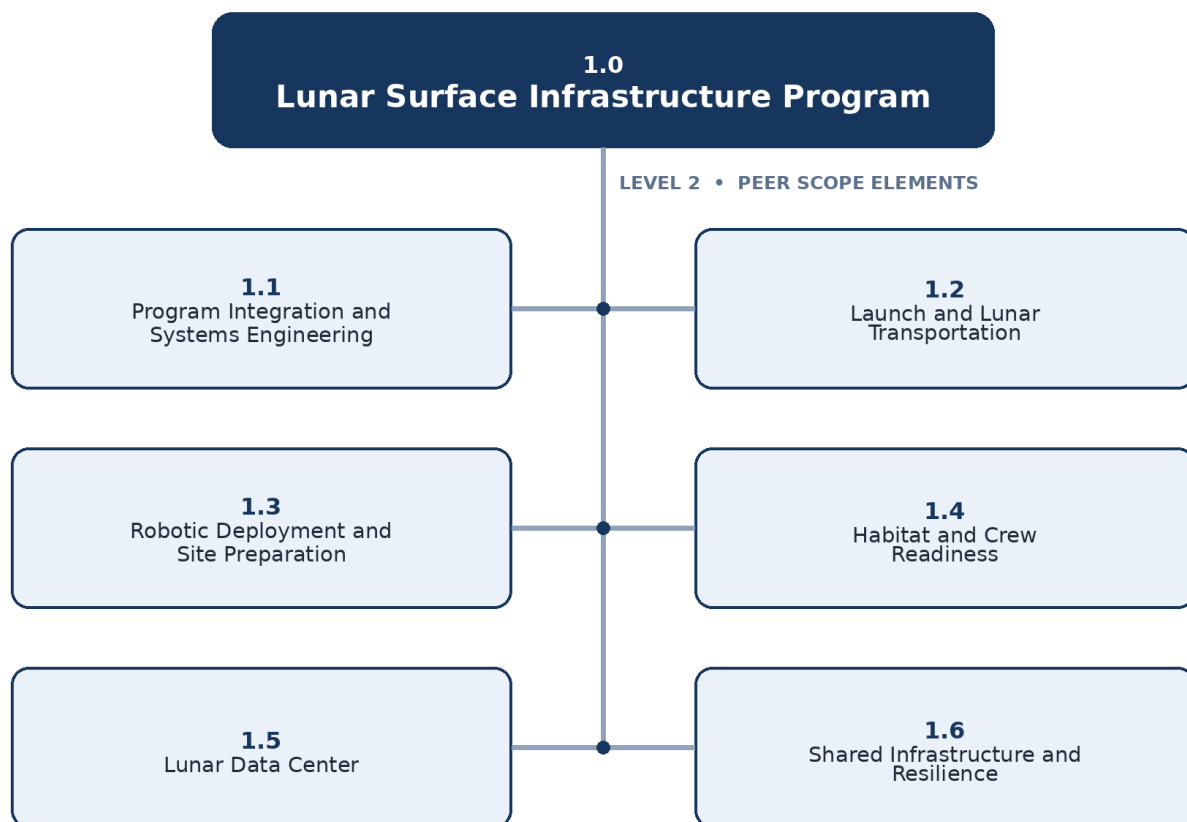


Figure 1. Level 1 program scope decomposed into six peer Level 2 elements.

WBS hierarchy

WBS	Level	Element	Primary deliverable
1.0	1	Lunar Surface Infrastructure Program	Integrated lunar demonstration planning baseline.
1.1	2	Program Integration and Systems Engineering	Controlled program baseline and integrated decision framework.
1.2	2	Launch and Lunar Transportation	Robotic, habitat, utility, and computing assets delivered to the lunar surface.
1.3	2	Robotic Deployment and Site Preparation	Surveyed, prepared, equipped, and remotely verified operating site.
1.4	2	Habitat and Crew Readiness	Operational habitat capable of supporting seven crew members.
1.5	2	Lunar Data Center	Commissioned computing capability within approved operating limits.
1.6	2	Shared Infrastructure and Resilience	Common utilities and recovery capabilities supporting all projects.

Decomposition rules

- The WBS is deliverable-oriented: each element represents a defined program outcome rather than an organization or calendar phase.
- Project-specific equipment and work remain within the applicable project element; services supporting multiple projects belong under 1.6.
- Program-level governance, integration, requirements, interfaces, configuration control, and readiness evidence belong under 1.1.
- Each lower-level element must map to one Level 2 parent and must not duplicate scope assigned elsewhere.
- Schedule activities, budgets, resources, risks, and performance measures will be assigned only after the WBS and dictionary are approved.

Baseline decision

Approve the Level 1 through Level 3 structure as the scope architecture for WBS Dictionary development and selective decomposition into Level 4 work packages.

Level 2 Element Definitions

Scope boundaries and completion conditions

1.1 Program Integration and Systems Engineering

Includes program governance, requirements and interface management, integrated schedule and resource planning, configuration control, risk and opportunity management, technical integration, verification planning, readiness reviews, and executive communication.

Completion condition: The integrated baseline is controlled, interfaces are defined, risks are managed, and decision-gate evidence is accepted.

1.2 Launch and Lunar Transportation

Includes payload integration, launch-service planning, trans-lunar transportation, lunar-orbit operations, descent and landing, payload separation, surface-delivery verification, and transportation contingencies. Development of a new launch vehicle is excluded.

Completion condition: Required assets are safely delivered, separated, located, and transferred to the appropriate lunar-surface project.

1.3 Robotic Deployment and Site Preparation

Includes robotic commissioning and communications, terrain and hazard surveys, site preparation and grading, utility and equipment placement, infrastructure inspection, repair support, and remote readiness verification.

Completion condition: The selected site satisfies approved readiness criteria before habitat deployment and crew launch.

1.4 Habitat and Crew Readiness

Includes habitat deployment and commissioning, pressure and atmospheric control, life support, water, food, consumables, waste management, medical capability, radiation protection, emergency response, crew communications, and safe-departure capability.

Completion condition: The habitat is verified to support seven crew members for a 45-day nominal mission with at least 60 days of critical provisions.

1.5 Lunar Data Center

Includes computing equipment, data storage and processing, dedicated power distribution, energy storage, heat rejection, communications, cybersecurity, autonomy, monitoring, load management, fault isolation, recovery, and maintenance capability.

Completion condition: Critical computing operates within approved power, thermal, communications, cybersecurity, maintenance, and recovery limits.

1.6 Shared Infrastructure and Resilience

Includes common surface power, shared communications and navigation, common thermal services, logistics, spares, dust mitigation, environmental and hazard monitoring, emergency power, cross-system fault response, contingency operations, and integrated recovery exercises.

Completion condition: Common services support the robotic, habitat, and data-center projects without compromising crew survival or safe departure.

1.1 Program Integration and Systems Engineering

Level 3 deliverable decomposition

Branch objective

Establish and control the integrated program baseline, technical architecture, decision framework, and readiness evidence across all projects.

Level 3 deliverables

WBS	Level 3 deliverable	Preliminary completion evidence
1.1.1	Program Governance and Decision Management	Governance cadence, delegated authorities, and decision records are approved and operating.
1.1.2	Requirements and Configuration Baseline	Program requirements and controlled configuration items are approved, traceable, and change-controlled.
1.1.3	Integrated Master Planning and Controls	Integrated scope, schedule, resources, and performance-measurement structure are baselined.
1.1.4	Systems Architecture and Interface Management	System architecture and cross-project interfaces are defined, allocated, and accepted.
1.1.5	Risk Opportunity and Issue Management	Program risks, opportunities, and issues have accountable owners, responses, and escalation paths.
1.1.6	Verification Validation and Readiness Assurance	Verification strategy, evidence requirements, and decision-gate criteria are approved.
1.1.7	Stakeholder and Executive Communications	Program status, decisions, and forecasts are communicated through an approved reporting structure.

Scope boundary

Project-specific design and execution remain within elements 1.2 through 1.6; this branch owns integration, governance, and program-level control.

Decomposition status: These elements define deliverable outcomes. The WBS Dictionary will establish exclusions, acceptance criteria, dependencies, assumptions, ownership, and risks before selected elements are authorized as work packages.

1.2 Launch and Lunar Transportation

Level 3 deliverable decomposition

Branch objective

Deliver robotic, habitat, utility, and computing assets from Earth through verified transfer to the lunar-surface projects.

Level 3 deliverables

WBS	Level 3 deliverable	Preliminary completion evidence
1.2.1	Mission Transportation Architecture	Transportation sequence, mission phases, vehicle roles, margins, and delivery interfaces are approved.
1.2.2	Payload Integration and Certification	Payloads satisfy mechanical, electrical, software, environmental, and safety integration requirements.
1.2.3	Launch Services and Ground Operations	Launch services, ground processing, range support, and launch-readiness evidence are accepted.
1.2.4	Trans Lunar and Lunar Orbit Operations	Transfer burns, navigation, communications, and lunar-orbit operations are planned and verified.
1.2.5	Descent Landing and Surface Delivery	Assets are landed within approved constraints, located, and transferred to surface operations.
1.2.6	Transportation Contingency and Recovery	Abort, delay, off-nominal delivery, and recovery responses are approved and exercised.

Scope boundary

This branch ends after safe surface delivery and transfer of custody; development of a new launch vehicle remains excluded.

Decomposition status: These elements define deliverable outcomes. The WBS Dictionary will establish exclusions, acceptance criteria, dependencies, assumptions, ownership, and risks before selected elements are authorized as work packages.

1.3 Robotic Deployment and Site Preparation

Level 3 deliverable decomposition

Branch objective

Use robotic systems to characterize, prepare, equip, inspect, and verify the operating site before crew arrival.

Level 3 deliverables

WBS	Level 3 deliverable	Preliminary completion evidence
1.3.1	Robotic Fleet and Payload Configuration	Robotic vehicles, tools, payloads, spares, and operating concepts are configured and accepted.
1.3.2	Lunar Site Survey and Hazard Characterization	Terrain, subsurface, illumination, dust, and hazard data support an approved site decision.
1.3.3	Surface Communications and Navigation Setup	Robotic command, telemetry, positioning, and navigation services meet operating criteria.
1.3.4	Site Preparation and Grading	Required operating areas and routes satisfy approved geometry, stability, and hazard limits.
1.3.5	Infrastructure Deployment and Assembly	Assigned equipment and infrastructure are positioned, connected, and functionally verified.
1.3.6	Inspection Maintenance and Remote Readiness Verification	The site passes inspection, exceptions are dispositioned, and remote readiness is approved.

Scope boundary

This branch owns robotic surface execution and site preparation; shared utilities delivered for multiple projects remain under 1.6.

Decomposition status: These elements define deliverable outcomes. The WBS Dictionary will establish exclusions, acceptance criteria, dependencies, assumptions, ownership, and risks before selected elements are authorized as work packages.

1.4 Habitat and Crew Readiness

Level 3 deliverable decomposition

Branch objective

Provide a verified habitat and operating capability for seven crew members during the planned surface mission.

Level 3 deliverables

WBS	Level 3 deliverable	Preliminary completion evidence
1.4.1	Habitat Deployment and Commissioning	The habitat is deployed, sealed, activated, and commissioned within approved limits.
1.4.2	Environmental Control and Life Support	Atmosphere, pressure, oxygen, carbon-dioxide removal, humidity, and water systems meet crew criteria.
1.4.3	Crew Consumables and Waste Management	Food, water, hygiene, consumables, storage, and waste provisions support the approved duration.
1.4.4	Radiation Dust and Thermal Protection	Crew exposure and habitat contamination remain within approved radiation, dust, and thermal limits.
1.4.5	Crew Health Safety and Emergency Response	Medical, fire, depressurization, shelter, rescue, and emergency procedures are verified.
1.4.6	Crew Communications Operations and Safe Departure	Crew communications, operating procedures, contingency return, and departure capability are ready.
1.4.7	Crew Mission Readiness and Training	Crew members demonstrate proficiency against approved mission and emergency scenarios.

Scope boundary

This branch owns crew-specific systems and readiness; common infrastructure serving multiple projects remains under 1.6.

Decomposition status: These elements define deliverable outcomes. The WBS Dictionary will establish exclusions, acceptance criteria, dependencies, assumptions, ownership, and risks before selected elements are authorized as work packages.

1.5 Lunar Data Center

Level 3 deliverable decomposition

Branch objective

Deliver and commission computing services that operate within approved power, thermal, communications, security, and recovery limits.

Level 3 deliverables

WBS	Level 3 deliverable	Preliminary completion evidence
1.5.1	Computing and Storage Architecture	Compute, storage, workload, redundancy, and capacity architecture satisfy approved service requirements.
1.5.2	Power Distribution and Energy Storage	Dedicated distribution, conversion, protection, and storage support required computing loads.
1.5.3	Thermal Control and Heat Rejection	Thermal transport and heat rejection maintain equipment within approved operating limits.
1.5.4	Communications and Data Transfer	Command, telemetry, data transfer, latency, and availability meet approved service criteria.
1.5.5	Cybersecurity and Access Control	Identity, access, segmentation, monitoring, data protection, and incident response controls are accepted.
1.5.6	Autonomous Monitoring and Fault Management	Monitoring, anomaly detection, isolation, load shedding, and automated recovery are verified.
1.5.7	Commissioning Performance Validation and Recovery	The data center passes performance, resilience, degraded-mode, and restoration demonstrations.

Scope boundary

This branch owns data-center-specific equipment and controls; common utilities serving the broader site remain under 1.6.

Decomposition status: These elements define deliverable outcomes. The WBS Dictionary will establish exclusions, acceptance criteria, dependencies, assumptions, ownership, and risks before selected elements are authorized as work packages.

1.6 Shared Infrastructure and Resilience

Level 3 deliverable decomposition

Branch objective

Provide common utilities, logistics, hazard controls, and recovery capability required by the robotic, habitat, and data-center projects.

Level 3 deliverables

WBS	Level 3 deliverable	Preliminary completion evidence
1.6.1	Common Surface Power	Shared generation, distribution, protection, storage interfaces, and priority rules are commissioned.
1.6.2	Shared Communications Navigation and Timing	Common communications, positioning, timing, and network services meet site requirements.
1.6.3	Common Thermal Services	Shared heat-transport or rejection services meet allocated loads and interface requirements.
1.6.4	Logistics Spares and Maintenance Support	Spares, tools, inventory, transport, maintenance data, and replenishment processes are ready.
1.6.5	Dust Mitigation and Environmental Monitoring	Site monitoring and dust prevention, detection, removal, and response capabilities are operational.
1.6.6	Emergency Power and Contingency Operations	Critical-load priorities, emergency power, safe states, and contingency procedures are verified.
1.6.7	Integrated Resilience and Recovery Demonstration	Cross-system disruption, isolation, degraded operation, and recovery scenarios are successfully demonstrated.

Scope boundary

Only services used by multiple projects belong here; project-dedicated equipment remains within its owning Level 2 element.

Decomposition status: These elements define deliverable outcomes. The WBS Dictionary will establish exclusions, acceptance criteria, dependencies, assumptions, ownership, and risks before selected elements are authorized as work packages.

Control Decisions and Next Step

Version 0.2 review and authorization

Revision summary

Version 0.2 retains the approved Level 1 and Level 2 architecture and adds 40 Level 3 deliverables across the six program elements. No Level 2 scope was removed or reassigned.

Approval decision

Approve the Level 3 decomposition as the working scope baseline for development of the WBS Dictionary. Approval confirms the scope structure; it does not authorize execution, funding, contractual commitments, or crewed mission operations.

Next authorized artifact

Develop the WBS Dictionary as a separate controlled artifact. Each Level 3 entry will define scope, exclusions, deliverables, acceptance criteria, dependencies, assumptions, accountable ownership, risks, and applicable decision gates. The dictionary will also identify which Level 3 elements require Level 4 work packages before scheduling and resource loading.

Management sequence

- Review the Level 3 structure for complete scope coverage and eliminate overlap between branches.
- Approve or revise deliverable names, boundaries, and preliminary completion evidence.
- Create the WBS Dictionary and assign accountable owners for development.
- Select elements requiring Level 4 work-package decomposition based on control needs and risk.
- Build the integrated master schedule from approved work packages and measurable activities.

Revision record: Version 0.2, 21 September 2026. Added Level 3 deliverables and preliminary completion evidence to all six Level 2 elements; Level 1 and Level 2 scope architecture remains unchanged.